
TECHNICAL SPECIFICATION

Developer Blog

Astro + MDX Static Site — Architecture & Implementation Spec v2.0

Document: Technical Specification v2.0

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Status: Draft — Review Required

Stack: Astro / MDX / Tailwind CSS / Vercel

This document supersedes the initial project brief. It provides the full architectural specification, data models, component inventory, build behavior, and acceptance criteria required for implementation.

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1. EXECUTIVE SUMMARY & SCOPE

This project is a personal developer blog. It is a statically generated content site authored by a single writer, served as pre-built HTML from a CDN. The architecture is intentionally simple: flat-file content, no database, no authentication, no dynamic server logic.

Scope Boundaries

- Static site generation from MDX content files
- Four route types: Home, Blog Index, Individual Post, Tag Filter
- Syntax-highlighted code blocks with copy-to-clipboard
- Estimated reading time per post
- SEO meta tags, OpenGraph, JSON-LD, sitemap, RSS feed
- GitHub Discussions-backed comments via Giscus
- Dark/light theme toggle with system preference detection
- Responsive layout (mobile-first, 3 breakpoints)
- Accessible markup meeting WCAG 2.1 AA

Explicitly Out of Scope

- User authentication or author login
- CMS admin panel or visual editor
- Newsletter subscription backend (use external service if needed)
- Search functionality (defer to future phase)
- Internationalization / multi-language support
- E-commerce or monetization features

2. TECHNOLOGY STACK — JUSTIFIED SELECTIONS

Each technology choice is listed with its rationale and the specific version to pin.

Layer	Technology	Version	Rationale
Framework	Astro	4.x	Zero-JS default output. Content collections API enforces schema.
Content	MDX	3.x	Markdown authoring with JSX escape hatch for interactive embeds.
Styling	Tailwind CSS	3.x	Utility-first. Purges unused CSS at build for minimal payload.
Syntax	Shiki	1.x	Built into Astro. VS Code grammar-based tokenizer. No runtime JS.
Comments	Giscus	latest	Maps to GitHub Discussions. No separate DB. Free. Themeable.
Media	Cloudinary	n/a	CDN-hosted images with on-the-fly transforms (resize, format).
Deploy	Vercel	n/a	Git-push deploys. Global edge CDN. Preview deploys on PRs.
Pkg Mgr	pnpm	9.x	Fast installs, strict dependency resolution, disk-efficient.

Rejected Alternatives

Next.js was considered but rejected: it defaults to client-side React hydration, which is unnecessary overhead for a content site with no dynamic UI. Gatsby was rejected due to its heavy plugin ecosystem and slower build times. Hugo was rejected because it lacks first-class JSX component support.

3. CONTENT DATA MODEL

Content is stored as .mdx files in [src/content/blog/](#). Each file's frontmatter is validated at build time by Astro's Content Collections schema. A build error is thrown if any required field is missing or malformed.

3.1 Frontmatter Schema

Field	Type	Required	Default	Description
title	string	Yes	—	Post title. Max 100 chars. Used in meta title.
slug	string	No	filename	URL slug. Decouples URL from filename.
description	string	Yes	—	Meta description. 50-160 chars for SEO.
publishDate	ISO 8601	Yes	—	Publication date. Used for sorting.
updatedAt	ISO 8601	No	null	Last update. Shown if newer than publishDate.
author	string	No	config default	Author display name.
tags	string[]	Yes	[]	Flat array of lowercase tag strings.
coverImage	URL string	No	null	Absolute URL to Cloudinary image.
coverImageAlt	string	Cond.	—	Required if coverImage is set. Alt text.
draft	boolean	No	false	If true: excluded from prod build entirely.
canonicalURL	URL string	No	null	For cross-posted content. Sets canonical link.

3.2 Schema Enforcement (Astro Config)

The schema must be defined in [src/content/config.ts](#) using Zod. Build fails on validation errors. This catches typos and missing fields before deployment, not after.

3.3 Tag Taxonomy

Tags are free-form lowercase strings (e.g., `react`, `system-design`, `devops`). No tag registry exists; tags are derived from the union of all post frontmatter at build time. Tags containing spaces must use hyphens. Maximum of 5 tags per post to avoid dilution.

4. ROUTING & PAGE ARCHITECTURE

Route	Source File	Behavior
/	src/pages/index.astro	Hero section with author bio. Grid of 5 most recent published posts.
/blog	src/pages/blog/index.astro	Chronological list. Paginated (10 posts/page). No infinite scroll.
/blog/[slug]	src/pages/blog/[...slug].astro	Dynamic route. Renders MDX body. Includes ToC, reading time, Giscus.
/tags	src/pages/tags/index.astro	Lists all tags with post count. Sorted alphabetically.
/tags/[tag]	src/pages/tags/[tag].astro	Filtered post list for a specific tag. Paginated.
/rss.xml	src/pages/rss.xml.ts	Generated RSS 2.0 feed. Published posts only.
/404	src/pages/404.astro	Custom 404 with navigation back to home.

4.1 Pagination Rules

- Page size: 10 posts per page.
- URL pattern: [/blog/page/2](#), [/blog/page/3](#), etc. Page 1 is [/blog](#) (no [/page/1](#)).
- Navigation: Previous / Next links. No page number jumps.
- Tag pages follow the same pagination: [/tags/react/page/2](#).

5. COMPONENT INVENTORY

All components are Astro components (.astro) unless they require client-side interactivity, in which case they use a framework island (Preact preferred for minimal bundle size).

Component	Type	Props / Behavior
BaseLayout	Astro	Wraps all pages. Includes head meta, nav, footer, theme script.
PostLayout	Astro	Extends BaseLayout. Adds article header, ToC, Giscus. Used by [slug].
Navbar	Astro	Site title, nav links (Home, Blog, Tags). Mobile hamburger.
PostCard	Astro	Displays cover image, title, date, reading time, tags. Used in lists.
TagBadge	Astro	Clickable pill linking to /tags/[tag]. Color derived from tag hash.
TableOfContents	Preact island	Sticky sidebar. Highlights active heading on scroll. client:idle.
ThemeToggle	Preact island	Sun/moon icon. Persists to localStorage. client:load.
CopyCodeButton	Preact island	Overlays code blocks. Copies content to clipboard. client:visible.
GiscusComments	Astro	Loads Giscus script. Maps to pathname. Lazy loaded.
Pagination	Astro	Prev/Next links. Receives currentPage, totalPages, basePath.

6. FEATURE SPECIFICATIONS

6.1 Syntax Highlighting

- Engine: Shiki (built into Astro via [@astrojs/mdx](#)).
- Theme: [github-dark-dimmed](#) (dark mode), [github-light](#) (light mode).
- Languages: All languages supported by VS Code grammars. Explicitly test: JavaScript, TypeScript, Python, Rust, Go, Bash, YAML, JSON, SQL, JSX/TSX, CSS, HTML, Diff.
- Line numbers: Disabled by default. Enabled per-block via [showLineNumbers](#) meta attribute.
- Line highlighting: Supported via [{1,3-5}](#) syntax in code fence meta.
- Copy button: Injected into every code block at top-right. Shows checkmark on success for 2 seconds.

6.2 Reading Time

- Calculation: [Math.ceil\(wordCount / 238\)](#). 238 WPM is the accepted average for technical content.
- Code blocks count at 0.4x weight (readers scan code, not read word-by-word).
- Display format: [X min read](#). Always rounded up. Minimum 1 min.
- Location: Under the post title, next to the publish date.

6.3 Table of Contents

- Generated from H2 and H3 headings in the MDX body.
- Rendered as a sticky sidebar on desktop (screen width over 1280px).
- Collapsed into an expandable section on mobile, placed above the article body.
- Active heading highlighted based on scroll position using IntersectionObserver.
- Smooth-scroll on click with scroll-margin-top accounting for navbar height.

6.4 Theme System

- Three states: Light, Dark, System. Default is System.
- Preference persisted to [localStorage](#) key [theme](#).
- Inline script in head prevents flash of wrong theme (FOWT) on page load.
- CSS custom properties for all colors. No hard-coded color values in components.
- Giscus theme updates reactively when blog theme changes.

6.5 Giscus Comments

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- Mapping strategy: [pathname](#) (maps each post URL to a Discussion thread).
 - Repository: Must be public. Discussions feature enabled.
 - Category: Create a dedicated [Blog Comments](#) category in GitHub Discussions.
 - Lazy loading: Giscus iframe loaded only when user scrolls to comments section.
 - Theme: Synced with blog theme (dark/light) via [postMessage](#).

7. BUILD PIPELINE & DRAFT BEHAVIOR

7.1 Build Environments

Behavior	Development (astro dev)	Production (astro build)
Draft posts	Visible. Marked with [DRAFT] badge.	Completely excluded. Not in HTML, RSS, sitemap, or tag counts.
Future-dated posts	Visible.	Excluded if publishDate is after build date.
Source maps	Enabled.	Disabled.
CSS	All utilities loaded.	Purged. Only used classes ship.
Image optimization	Skipped for speed.	Full optimization: WebP/AVIF, responsive srcset.

7.2 Vercel Deployment

- Trigger: Push to [main](#) branch.
- Preview deploys: Every pull request gets a unique preview URL.
- Build command: [pnpm build](#).
- Output: [dist/](#) directory. Pure static files.
- Environment variables: [SITE_URL](#), [GISCUS_REPO](#), [GISCUS_REPO_ID](#), [GISCUS_CATEGORY_ID](#).

8. PERFORMANCE BUDGETS

These are hard limits, not guidelines. A pull request that regresses any metric below these thresholds must not be merged.

Metric	Target	Tool
Lighthouse Performance	over 95	Lighthouse CI in GitHub Actions
First Contentful Paint	under 1.0s	WebPageTest (3G throttle)
Largest Contentful Paint	under 1.5s	WebPageTest
Cumulative Layout Shift	under 0.05	Lighthouse
Total page weight (no images)	under 50 KB	Bundlewatch
JavaScript shipped	under 10 KB	Build output analysis
CSS shipped	under 15 KB	Build output analysis
Time to Interactive	under 1.5s	Lighthouse

9. ACCESSIBILITY REQUIREMENTS

Target: WCAG 2.1 Level AA compliance.

- All images must have meaningful `alt` text. Decorative images use `alt=""` with `role="presentation"`.
- Color contrast ratio: minimum 4.5:1 for body text, 3:1 for large text. Both themes.
- Full keyboard navigation: all interactive elements focusable and operable. Visible focus rings.
- Skip-to-content link as the first focusable element on every page.
- Semantic HTML: `article`, `nav`, `main`, `aside`, `header`, `footer` used correctly.
- Heading hierarchy: no skipped levels. One H1 per page.
- Code blocks: scrollable regions are keyboard-accessible.
- Reduced motion: respect `prefers-reduced-motion` media query. Disable animations.
- Language attribute set on HTML element.

10. SEO SPECIFICATION

10.1 Meta Tags (Every Page)

Tag	Source	Notes
<code>title</code>	<code>frontmatter.title</code> + site name	Format: Post Title Blog Name
<code>meta.description</code>	<code>frontmatter.description</code>	50-160 characters
<code>canonical</code>	<code>frontmatter.canonicalURL</code> or self	Prevents duplicate content
<code>og:title</code>	Same as title	—
<code>og:description</code>	Same as description	—
<code>og:image</code>	<code>coverImage</code> or site default	1200x630 recommended
<code>og:type</code>	article (posts), website (other)	—
<code>og:url</code>	Full canonical URL	—
<code>twitter:card</code>	<code>summary_large_image</code>	—
<code>article:published_time</code>	<code>publishDate</code>	ISO 8601
<code>article:modified_time</code>	<code>updatedAt</code>	If present
<code>article:tag</code>	<code>tags</code> array	One per tag

10.2 Structured Data (JSON-LD)

Every blog post page must include a [BlogPosting](#) schema in a script tag of type `application/ld+json`. Fields: `headline`, `description`, `datePublished`, `dateModified`, `author` (Person), `image`, `url`, `mainEntityOfPage`.

10.3 Sitemap

Generated at build via [@astrojs/sitemap](#). Excludes drafts and 404. Includes `lastmod` from `updatedAt` or `publishDate`. Submitted to Google Search Console.

10.4 RSS Feed

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- Generated at </rss.xml> via [@astrojs/rss](#).
 - Contains: title, description, link, pubDate, content (full HTML body).
 - Published posts only. Sorted by publishDate descending.
 - Feed auto-discovery link tag in the head of every page.

11. ANALYTICS & MONITORING

Privacy-respecting analytics only. No cookies, no tracking scripts that require consent banners.

- Recommended: Plausible Analytics or Fathom (privacy-first, GDPR-compliant, no cookie banner needed).
- Alternative: Vercel Web Analytics (built-in, zero-config).
- Track: page views, referrers, top pages, country. Nothing else.
- No Google Analytics. The consent banner requirement degrades UX and contradicts the minimal design.

12. SECURITY CONSIDERATIONS

- Content Security Policy headers configured in [vercel.json](#). Allowlist for Giscus iframe, Cloudinary images, Plausible script.
- All external resources loaded over HTTPS.
- No user input accepted server-side (static site). XSS surface limited to Giscus iframe (sandboxed).
- Dependencies audited with [pnpm audit](#) in CI. Build fails on critical vulnerabilities.
- Subresource Integrity (SRI) hashes on any third-party scripts.
- [X-Frame-Options: DENY](#) to prevent clickjacking.
- [Referrer-Policy: strict-origin-when-cross-origin](#).

13. FUTURE CONSIDERATIONS

These items are explicitly deferred. They are documented here so they inform current architectural decisions without bloating the initial scope.

Feature	Why Deferred	Architectural Impact
Full-text search	Low ROI until over 50 posts	May need Pagefind or Algolia integration. No impact on current schema.
Series / multi-part posts	Need content first	Add optional series and seriesOrder fields to frontmatter.
Newsletter integration	Requires audience first	Embed form via ConvertKit/Buttondown. No backend change.
View count display	Requires analytics API	Vercel Analytics API or Plausible. Read-only fetch at build or client.
Related posts	Needs tag overlap algorithm	Compute at build time. Tag intersection scoring.
Guest author profiles	Solo blog for now	Evolve author from string to reference type in schema.

14. ACCEPTANCE CRITERIA

The project is considered complete when all of the following are verified:

1. A new .mdx file with valid frontmatter, committed to main, appears as a published post on the live site within 3 minutes.
2. A post with `draft: true` does not appear anywhere in production: not in HTML, RSS, sitemap, tag counts, or related posts.
3. All code blocks render with correct syntax highlighting and a functional copy button.
4. The reading time displayed matches the formula: `ceil(wordCount / 238)`.
5. Every post page has valid OpenGraph tags. Verified via LinkedIn Post Inspector and Twitter Card Validator.
6. The RSS feed validates against the W3C Feed Validation Service with zero errors.
7. Lighthouse Performance score is 95 or above on both a post page and the blog index (mobile throttling).
8. All pages pass axe-core automated accessibility checks with zero violations.
9. Theme toggle works without FOWT. Persists across page navigations and browser sessions.
10. Giscus comments load on scroll, create a new Discussion thread on first comment, and sync theme with the blog.
11. Pagination works correctly on blog index and tag pages with 10+ posts.
12. The 404 page renders for unknown routes and provides navigation back to home.
13. The site is fully functional on latest Chrome, Firefox, Safari, and Edge.
14. Mobile layout is usable at 375px viewport width (iPhone SE).

END OF SPECIFICATION

This document should be treated as a living spec. Update it as decisions are made during implementation.